EBRAR ACAR

Software Engineering Student

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SUMMARY

Senior Software Engineering student specializing in autonomous multi-agent systems, swarm robotics, and edge AI. Experienced in developing mission software for multi-UAV swarm systems and multi-agent reinforcement learning based traffic signal control architectures, with strong practical skills in ROS, Python, and C++. Seeking to apply this experience to swarm autonomy and edge intelligence challenges in the space sector.

EDUCATION

Software Engineering **2023-Present**

Konya Technical University - GPA: 3.83

EXPERIENCE

MOSAŞ Group - Artificial Intelligence Intern **June - August 2026**

- Worked on and tested multi-agent reinforcement learning architectures that coordinate traffic signals to reduce congestion across different simulators and traffic networks, including a real intersection network, with results being prepared as a research paper.

PROJECTS

Task Allocation and Autonomous Coordination for a ROS-Based UAV Swarm System - Capstone Project

- **Carrier UAV:** Developed a Brute Force based route planning software in C++ that guides 4 mini UAVs to their target zones in the shortest time.
- **Human Tracking:** Developed a stable tracking system combining a Python-based image processing node (MobileNet-SSD object detection, SORT/Kalman Filter based ID tracking, and uniform color based authorization) with C++-based PID control (forward-backward, lateral, yaw) and LIDAR integration.
- **Area Scanning:** Implemented autonomous area scanning with MobileNet-SSD based object detection in a Python image processing node, C++-based PID controlled scanning with user-defined parameters, and a safe payload drop after detection.
- **Object Detection:** Designed a system using OpenCV that performs area scanning on a fixed route with AruCo marker detection, transmitting detected coordinates in real time to the kamikaze UAV.
- **Kamikaze:** Developed autonomous flight software in C++ that simulates precise dive maneuvers toward target coordinates using PID control.
- The swarm coordination architecture is currently under development.

Multi-Agent Reinforcement Learning Based Traffic Signal Control Benchmark - Internship Project

- Benchmarked multi-agent RL methods (CoLight, MPLight) against non-coordinating baselines (Fixed-time, MaxPressure) on the LibSignal framework, using a real 4×4 intersection network to measure the contribution of inter-agent coordination to reducing congestion.
- Repeated the study across two simulators (CityFlow and SUMO) and two traffic distributions (homogeneous and heterogeneous vehicles) to examine the consistency of coordination across environments.
- Ported the CityFlow-only DHLight model to SUMO, integrated it into the LibSignal framework myself, and successfully tested it in the benchmark.

IoT-Based Fully Autonomous UAV Project - 2025 Teknofest International UAV Competition

- Developed ROS-based autonomous route calculation and payload drop software in Python, and set up a cloud-based MQTT communication infrastructure linking Raspberry Pi units on different networks to the ground station computer.

rotAY - Hybrid Autonomous Route Optimization System for Lunar Rovers (TUA Astro Hackathon Project)

- Served as team lead in developing a hybrid autonomous route planning system for lunar rovers based on A* and Genetic Algorithm, working on NASA LRO elevation data (MoonPlanBench).
- Designed a profile-based cost function that simultaneously optimizes distance, safety clearance, and terrain roughness (Fast, Balanced, and Safe mission scenarios), within a two-layer architecture: global waypoint optimization with Genetic Algorithm before the mission and real-time local A* re-planning for dynamic obstacles.
- Developed the simulation environment and managed testing using Python, NumPy, SciPy, and scikit-image.

Turtlebot3 Cleaner

- Developed QR-code-based cleaning robot software using ROS Noetic, including an OpenCV-based ROS node for QR code detection.
- Performed environment mapping via RViz using SLAM algorithms and implemented autonomous navigation using move_base.
- Designed the system with custom launch files for a modular, manageable structure.

TECHNICAL SKILLS

- **Programming Languages:** C++ • Python
- **Embedded Systems & Robotics:** Raspberry Pi • Jetson Nano • ROS • Gazebo
- **Communication & Networking:** TCP/IP • MQTT
- **Software Design:** SOLID Principles • Clean Code • OOP • Design Patterns
- **Tools & Other:** Docker • Git

LANGUAGES

- **English** - C1 (YÖKDİL: 90)

CERTIFICATIONS & ACHIEVEMENTS

- **International Astronautical Congress (IAC) 2026:** Presented research on swarm-based positioning for Mars aerial exploration.
- **3rd Place, Konya Technical University 10th Project Market:** “Autonomous Multi-UAV System V2”, a team project whose software I developed as my capstone project.
- **Defense Industry Academy 401 Training (Savunma Sanayii Akademi, 2026):** Defense industry projects, systems engineering, and project management.
- **Autonomous Driving Technologies Program - Basic Training (Ministry of Industry and Technology, 2026):** Data processing and autonomous system integration.